

Gokul Ram Kannan

AI/ML Researcher | Multimodal AI & Explainable Learning | Applied Machine Learning

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Education

B.Tech in Computer Science and Engineering (AI and ML), VIT Chennai Aug 2023 – Aug 2027
– CGPA: 8.8/10.0

Research Experience

- **Research Engineer Intern**, Centre for Neuroinformatics (CNI), VIT Chennai (Hybrid) May 2026 – Jul 2026
CNI-QNeuro-SUNY Binghamton Research Program
 - Developed a **Cognitive Grammar Learning Framework** converting multimodal ECG + EDA into a **5-state symbolic vocabulary** via GMM/BIC and modeling state transitions with a **BiGRU**; integrated **CL-Drive + CLARE** into 637 synchronized 10-s windows with **629 validated samples**.
 - Engineered normalization, **latent-state discovery**, **grammar induction**, and **sequence-learning** pipelines; evaluated **34 participants** via LOSO and cross-dataset transfer for participant-, state-, transition-levels.
 - Built **explainability**, **reliability**, and **statistical validation** using Gradient Saliency, Integrated Gradients, ECE, PCA, BCa bootstrap, Shapiro-Wilk, jackknife, and grammar-similarity metrics.
- **Research Intern**, School of Computer Science and Engineering, VIT Chennai (Hybrid) May 2025 – Jul 2025
 - Developed **HEADER-Net**, a hierarchical speech-emotion disentanglement framework using frozen **emotion2vec+** representations, separating **stable/dynamic emotion** and **speaker/content nuisance**.
 - Engineered **temporal attention**, **Bi-LSTM**, **temporal-difference modeling**, **cross-level agreement**, **VAE-style purification**, **prototype learning**, **GRL adversaries**, **orthogonal disentanglement**, and **cross-reconstruction** under an **8-component multi-task loss**.
 - Evaluated **8,498 utterances / 115 speakers** with speaker-disjoint and cross-dataset protocols; achieved **0.8927 Macro-F1**, **0.0112 speaker Macro-F1**, and **0.4427 cross-dataset Macro-F1**, with ablations confirming core architectural contributions.

Selected Publications

Google Scholar

- **Multimodal Data Fusion for Compatibility Reasoning**, *Array, Elsevier, 2026 (I.F. 5.3)* **Accepted | First Author**
- **Feature Fusion and XAI for Tamil Speech Emotion Recognition**, *IEEE 7th ICIRCA, 2026* **First Author**
- **Deep Learning and XAI for Customer Sentiment Analysis**, *IEEE 4th ICUIS, 2024* **First Author | Cited by 7**
- **NLP for Protein Secondary Structure Prediction**, *Results in Engineering, Elsevier, 2024 (I.F. 9.4)* **Cited by 25**

Selected Research Projects

- Temporal Machine Learning for Multiclass Solar Flare Forecasting** May 2026 - Sep 2026
- Developed a **GOES-16 XRS multiclass forecasting framework** over **420,017 observations** with **40+ temporal features** spanning lagged flux, rolling statistics, flux differences, logarithmic transforms, and cyclic encodings; applied **Temporal-SMOTE (T-SMOTE)** for severe class imbalance.
 - Benchmarked **12 ML, ensemble, and recurrent models** under **chronological distribution shift** (2021–2022 → 2023–2024), including **LightGBM, XGBoost, Random Forest, CatBoost, Voting, Stacking, BiLSTM, and GRU**, using **Macro-F1, Macro-TSS, ROC-AUC, and class-wise evaluation**.
 - Performed **T-SMOTE ablation and rare-class analysis**, improving **M-class F1 from 0.274 to 0.408** and yielding non-zero **X-class F1**; quantified **temporal distribution-shift** effects and the limits of **XRS-only flare forecasting**.
- DriveGraph: Explainable AI Autonomous Driving Simulator** GitHub Mar 2026 – May 2026
- Engineered an **Explainable AI autonomous-driving simulator** combining **OpenStreetMap road graphs**, **kinematic dynamics**, **15 LiDAR-like 360° beams**, **semantic traffic sensing**, and a **sparse graph neural controller** with real-time decision visualization.
 - Evolved **neural topology and weights** across **50 policies × 100 generations × 10 episodes**, optimizing progress, safety, rule compliance, and comfort; expanded networks **25→39 nodes** and **45→78 connections** under held-out-map/seed evaluation with **TTC safety overrides**.
 - Achieved **89.7% ± 2.9% success**, **4.9% ± 1.2% collision**, and **94.2% ± 1.8% rule compliance**; reduced collisions by **44%** against the heuristic baseline and outperformed the fixed-topology NN.

Industry Experience

- **AI Engineer Intern**, Perartral Technologies India Pvt. Ltd., Chennai (On-Site) May 2026 – Present (Expected)
 - Engineered a **CV + multimodal-LLM floor-plan pipeline** using **EasyOCR, OpenCV, YOLO, Hough, and Gemini** to construct a validated **Building Semantic Model** across **8 engineering-agent contexts**.
 - Developed the **HVAC intelligence layer** over geometry, thermal zones, openings, walls, adjacency, and airflow topology, generating structured context for downstream engineering reasoning.
 - Built a **schema-aware NL-to-SQL WMS assistant** using **FastAPI, LlamaIndex, BGE-small, and Llama-3-8B**, implementing intent detection, entity matching, retrieval, SQL generation/validation/execution, and session management over a **4-sheet schema**.
- **Python Developer Intern**, KRG Technologies Inc., Chennai (On-site) Jun 2025 – Jul 2025
 - Engineered an **AI resume-intelligence pipeline** validated on **2,000 resumes** across **7+ formats**, combining native extraction and vision OCR with **95–99% core-field extraction**.
 - Built **25-field LLM-based extraction** with Gemini, JSON/schema validation, normalization, and **Flask APIs**; integrated document-processing pipelines for automated resume analysis.
 - Developed **React + Flask** workflows for individual/bulk processing and recruiter search across **20+ candidate attributes**, reducing manual screening overhead by **85–90%**.

Funded Research Project

Project Lead — VIT Chennai (On-Site) Aug 2025 – Present

Next-Gen Irrigation: Leveraging Machine Learning for Precision Water Delivery at VIT

- Led a **14-member multidisciplinary team** across 8 workstreams to deploy **10 Raspberry Pi IoT units** with 7 sensing channels each; built **3D-printed enclosures**, 15-minute acquisition, MQTT/JSON telemetry, and centralized aggregation, generating **604,800 observations** over 90 days.
- Engineered the **Raspberry Pi → MQTT → AWS IoT Core → EC2 → RDS/S3 → Flask** stack; implemented **JWT-secured APIs**, Lambda automation, and CloudWatch monitoring.
- Benchmarked **deep-learning architectures** across **6+ metrics** for multi-day sensor forecasting; deployed predictions for **automated irrigation valve control**, yielding **4 Indian patent applications**.

Open Source Contributions

GirlScript Summer of Code (GSSoC) 2026

- Ranked **#80 globally** among **43,587 contributors (Top 1%)** and **2nd at VIT Chennai**, with **179 merged PRs** across **7 repositories** spanning AI/Agents and Open Source.
- Contributed to production **LLM, RAG, and AI-agent systems** across **Next.js, Python, FastAPI, and Node.js**, implementing orchestration, structured-output validation, persistence, SSE, authenticated APIs, rate limiting, concurrency control, and security validation.

Skills & Competencies

Languages: Python, C++, Java, SQL, JavaScript, TypeScript

AI/ML: ML, DL, NLP, LLMs, RAG, CV, Speech, XAI, Multimodal Learning, Representation Learning

Methods: Self-Supervised Learning, Transfer Learning, Contrastive Learning, Adversarial Learning, Feature Fusion, Temporal Modeling, Domain Generalization, DS Evaluation, Ablation, Statistical Validation

Architectures: CNN, BiLSTM, BiGRU, GRU, GNN, VAE, AE, Attention, Ensembles

XAI/Stats: SHAP, LIME, IG, GS, PCA, t-SNE, Bootstrap, Jackknife, Hypothesis Testing, Calibration

Frameworks/Tools: PyTorch, TensorFlow, Scikit-learn, LlamaIndex, LangChain, FastAPI, Flask, OpenCV, YOLO

Cloud/Systems: AWS (EC2, S3, Lambda, IoT Core, RDS, CloudWatch), PostgreSQL, Supabase, ChromaDB, MQTT

Patents

5 published Indian patents (2023–2026)

Inventor | AI, IoT, Smart Agriculture, Affective Computing

Certifications

- **AWS Certified Solutions Architect – Associate (SAA-C03)** — AWS (Jul 2026) [Link](#)
- **Oracle Cloud Infrastructure Certified Generative AI Professional (1Z0-1127-25)** — Oracle (Jun 2026) [Link](#)
- **Oracle Cloud Infrastructure Certified AI Foundations Associate (1Z0-1122-25)** — Oracle (Jun 2026) [Link](#)
- **AWS Certified Cloud Practitioner (CLF-C02)** — AWS (Jan 2024) [Link](#)
- **Certified Associate Python Programmer (PCAP-31-03)** — Python Institute (Jul 2023) [Link](#)